17/08/26

Signal Processing

- Recap: Memoryless and linear systems
- Time invariant, causal, stable systems
- Linear and time-invariant systems
- Quiz 1
- Motivation: $y[n] = T\{x[n]\}$

  $T\{\ \}$: map from $\underbrace{x[n]}_{i/p}$ to $\underbrace{y[n]}_{o/p}$

- Time invariant systems: A system $T\{\ \}$ is time invariant if for all n_0, the output of the system to a "delayed" input $x[n-n_0]$ is equal to the delayed output $y[n-n_0]$. In other words if $x_1[n] = x[n-n_0]$, then $T\{\ \}$ is time invariant if $y_1[n] = T\{x_1[n]\} = T\{x[n-n_0]\} = y[n-n_0]$ (where $y[n] = T\{x[n]\}$). for all n_0.

Ex: Accumulator system: $y[n] = \sum_{k=-\infty}^{n} x[k]$

$$y_1[n] = \sum_{k=-\infty}^{n} x_1[k]$$

$$= \sum_{k=-\infty}^{n} x[k-n_0] \quad \text{(now, change of variable)} \quad k' = k - n_0$$

$$= \sum_{k'=-\infty}^{n-n_0} x[k'] \quad \text{— (1)}$$

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

$$y[n-n_0] = \sum_{k=-\infty}^{n-n_0} x[k] \quad \text{— (2)}$$